MONEY DURATION AND PRICE VALUE OF A BASIS POINT

Learning Outcome

- define, calculate, and interpret modified duration, money duration, and the price value of a basis point (PVBP)

Modified duration is used to measure the *percentage price change* of a bond given a change in its yield-to-maturity. A related statistic is **money duration**. The money duration of a bond is a measure of the price change in *currency units*. The money duration can be stated per 100 of par value or in terms of the actual position size. In the United States, money duration is commonly called "dollar duration."

Money duration (*MoneyDur*) is the product of the annualized modified duration and the full price (PV^{Full}) of the bond, in either percent of par or the currency value of the position.

$$MoneyDur = AnnModDur \times PV^{Full}. \quad 6$$

The estimated change in the bond price in currency units is very similar to Equation 4. The difference is that for a given change in the annual yield-to-maturity ($\Delta Yield$), modified duration estimates the percentage price change while money duration estimates the change in currency units.

$$\Delta PV^{Full} \approx -MoneyDur \times \Delta Yield \quad 7$$

BRWA's five-year, 3.2% semiannual coupon bond, maturing 15 October 2030, is priced to yield 3.2%. Assume it is purchased by an institutional investor for settlement on 11 December 2025, so 57 days into the first coupon period (t/T = 57/180).

The full price of the bond is 100.504 per 100 of par value, and the annualized modified duration is 4.43092. Suppose the investor has a position in the bond for a par value of USD100,000,000. Consequently,

- the market value of the investment is USD100,503,921, and
- the money duration of the investment is USD445,324,719 (= 4.43092 × USD100,503,921).

Therefore, if the yield-to-maturity rises by 100 bps—from 3.20% to 4.20%—the expected loss is approximately USD4,453,247 (= –USD445,324,719 × 0.0100).

A similar measure is the **price value of a basis point (PVBP)**, an estimate of the change in the full price of a bond given a 1 bp change in its yield-to-maturity. The PVBP can be calculated using a formula like that for approximating modified duration. Equation 8 is the formula for the PVBP. PV_- and PV_+ are the full prices calculated by decreasing and increasing the yield-to-maturity by 1 bp.

$$\text{PVBP} = \frac{(PV_-) - (PV_+)}{2}. \quad 8$$

The PVBP is also called the "PV01," standing for the "price value of an 01" or "present value of an 01," where "01" means 1 bp. In the United States, it is commonly called the "DV01" for the "dollar value" of 1 bp. The PVBP is particularly useful for bonds for which future cash flows are uncertain, such as callable bonds. A related statistic called a "basis point value" (BPV) is simply the money duration times 0.0001 (1 bp).

3.2% semiannual coupon bond, maturing 15 October 2030 and priced to yield 3.2%, purchased by the institutional investor for settlement on 11 December 2025, so 57 days into the first coupon period (t/T = 57/180).

To determine the PVBP of this bond, calculate the new prices by increasing and decreasing the yield-to-maturity. First, increase the yield by 1 bp (0.01%), from 3.20% to 3.21%, to solve for a PV_+ of 100.459400.

$$PV_+ = \left[\frac{1.60}{(1.01605)^1} + \ldots + \frac{101.60}{(1.01605)^{10}}\right] \times (1.01605)^{57/180} = 100.459400.$$

Then, decrease the yield-to-maturity by 1 bp, from 3.20% to 3.19%, to solve for a PV_- of 100.548465.

$$PV_- = \left[\frac{1.60}{(1.01595)^1} + \ldots + \frac{101.60}{(1.01595)^{10}}\right] \times (1.01595)^{57/180} = 100.548465.$$

The PVBP is obtained by substituting these results into Equation 8.

$$\text{PVBP} = \frac{100.548465 - 100.459400}{2} = 0.044532.$$

Exhibit 3 provides a summary of the various yield duration statistics. To recap, they measure the instantaneous change in a bond's price for a given change in its own yield-to-maturity, assuming all other variables (especially, time-to-maturity) are unchanged and the underlying cash flows are certain.

Exhibit 3:

Summary of Yield-Based Duration Statistics

Measure	Calculation	Use
Macaulay duration	Average time to receipt of promised cash flows, weighted by shares of the full price corresponding to each promised future cash flow	Holding period that would balance reinvestment and price risks for an investor
Modified duration	First derivative of price with respect to yield; Macaulay duration divided by 1 + yield per period	Estimate the percentage price change for a bond given a change in its yield-to-maturity
Money duration	Modified duration multiplied by full price of bond or bond position	Estimate price change in bond investment for a given yield change
Price value of a basis point	Difference in price of a 1 bp yield decrease and a 1 bp yield increase, divided by 2	Estimate of the change in the bond price given a 1 bp change in the yield-to-maturity

Yield Duration of Zero-Coupon and Perpetual Bonds

Because zero-coupon bonds have a single payment, the face value at maturity, the present weighted-average time to receipt of cash flows is the same as the time-to-maturity because that single cash flow has a present value weight of 1.0. Therefore, the Macaulay duration of a zero-coupon bond is its time-to-maturity, and modified duration is its time-to-maturity divided by 1 plus its yield.

A perpetuity or perpetual bond is a bond that does not mature, so there is no face or maturity value received at time T. The investor receives a fixed coupon payment forever unless the bond is called. Non-callable perpetuities are rare, but they have an interesting Macaulay duration:

$$MacDur = (1 + r)/r. \quad 9$$

Duration of Floating-Rate Notes and Loans

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payment amounts are reset to reflect changes in the MRR. Therefore, interest rate risk arises only *between* reset dates, because at the next reset date, coupon payments will adjust to the new MRR. Therefore, the Macaulay duration for a floating-rate note or bond is simply the fraction of a period remaining until the next reset date:

$$MacDur_{Floating} = \frac{(T-t)}{T}. \qquad 10$$

If there are 180 days in the coupon period and 57 days have passed since the last coupon, the Macaulay duration is

$$MacDur_{Floating} = \frac{(180-57)}{180} = 0.683333.$$

Floating-rate instruments typically have very low duration because coupon periods are typically less than six months in length. As a result, they are commonly used by investors to reduce duration in fixed-income portfolios.

Knowledge Check

An analyst is comparing the interest rate risk of the BRWA 3.2% bond maturing 15 October 2030 to a floating-rate note (FRN) that also matures 15 October 2030 and has its coupon reset semiannually.

At the last reset, 15 October 2027, the coupon rate of the FRN was 3.2%. The BRWA bond was priced to yield 3.2% at that same date.

1. Why would you expect the FRN to have a lower duration than the BRWA fixed-rate bond?

Hide Solution

Solution:

A floating-rate instrument has a Macaulay duration that reflects the time to the next coupon reset date, while a fixed-rate instrument has a Macaulay duration that primarily reflects time-to-maturity but also its coupon rate, yield, and fraction of the coupon period that has passed. The maximum Macaulay duration the FRN would have is six months, while the BRWA bond has a Macaulay duration slightly lower than its maturity (five years).

2. Complete the following table:

	Macaulay duration	
	BRWA	FRN
15-Oct-2027		
15-Nov-2027		
15-Dec-2027		
15-Jan-2028		
15-Feb-2028		
15-Mar-2028		
15-Apr-2028		

Hide Solution

Solution:

	Macaulay duration	
	BRWA	FRN
15-Oct-2027	2.8843	0.0000

15-Nov-2027	2.8002	0.8306
15-Dec-2027	2.7169	0.6667
15-Jan-2028	2.6336	0.4973
15-Feb-2028	2.5502	0.3279
15-Mar-2028	2.4669	0.1694
15-Apr-2028	2.4221	0.0000

For example, for 15 December 2027, the number of days since the next coupon is 61 days. The number of days in the coupon period is 183, calculated using Microsoft Excel or Google Sheets as

= COUPDAYS(DATE(2027,12,15),DATE(2028,4,15),2,1).

Therefore, the Macaulay duration is (183 – 61)÷183 = 0.6667.

Question Set

Deepak Chowdhury is evaluating a USD30 million investment in par value terms in the BRWA, Australian, and Romanian bonds. Assume all bonds are denominated in the same currency and that all three bonds were issued on 15 October 2025.

Chowdhury is evaluating the bonds on 15 January 2026, 92 days into the first coupon period. Chowdhury has compiled the following data.

Characteristic	BRWA	Australian	Romanian
Coupon	3.200%	0.000%	4.625%
Coupon frequency	2	1	1
Yield-to-maturity	3.200%	1.000%	4.250%
Maturity	15 October 2030	15 October 2026	15 October 2055
Full price	100.815	99.262	107.429
Ann. Macaulay duration	4.405	0.744	16.939
Ann. modified duration	4.335	0.737	16.248

As part of his analysis, Chowdhury is evaluating the potential change in the prices of the bond investments for 1 bp and 100 bp changes in yields.

1. Complete the following table:

	Effect on a USD30 million investment		
	BRWA	Australian	Romanian
Price value per basis point			
Market value of investment (USD)			
Money duration (percent of par)			
Expected loss from a 100 bp YTM increase (USD)			

Hide Solution

Solution:

	Effect on a USD30 million investment		
	BRWA	Australia	Romania
Price value per basis point	0.044	0.007	0.175
Market value of investment (USD)	30,244,381	29,778,597	32,228,627

	BRWA	Australia	Romania
Money duration (percent of par)	437.054	73.163	1745.518
Expected loss from a 100 bp YTM increase (USD)	–1,311,163	–219,490	–5,236,553

The calculations are shown in tabular form. Please refer to the candidate learning ecosystem online for a spreadsheet in the downloadable Microsoft Excel workbook. Note that while solutions are rounded to three decimal places, other quantities are unrounded but shown only to three decimal places for presentation purposes.

BRWA Bond: Full price, assuming 1 bp YTM increase		BRWA Bond: Full price, assuming 1 bp YTM decrease	
YTM (periodic)	0.01605	**YTM (periodic)**	0.01595

Period	Time to Receipt	Cash Flow	PV+	Period	Time to Receipt	Cash Flow	PV-
1	0.489	1.600	1.588	1	0.489	1.600	1.588
2	1.489	1.600	1.563	2	1.489	1.600	1.563
3	2.489	1.600	1.538	3	2.489	1.600	1.538
4	3.489	1.600	1.514	4	3.489	1.600	1.514
5	4.489	1.600	1.490	5	4.489	1.600	1.490
6	5.489	1.600	1.466	6	5.489	1.600	1.467
7	6.489	1.600	1.443	7	6.489	1.600	1.444
8	7.489	1.600	1.420	8	7.489	1.600	1.421
9	8.489	1.600	1.398	9	8.489	1.600	1.399
10	9.489	101.600	87.353	10	9.489	101.600	87.435
			100.771				100.858

BRWA Bond			
PV+	100.771	Full price	100.815
PV-	100.858	Par	30,000,000
PVBP	0.044	Market value	30,244,381
AnnModDur	4.335	MoneyDur	437.054
Full price	100.815	MV	30,244,381
MoneyDur	437.054	ChgYield	0.01
		Exp. Loss	(1,311,163)

Australian Bond: Full price, assuming 1 bp YTM increase		Australian Bond: Full price, assuming 1 bp YTM decrease	
YTM (periodic)	0.0101	**YTM (periodic)**	0.0099

Period	Time to Receipt	Cash Flow	PV+	Period	Time to Receipt	Cash Flow	PV-
1	0.744	100.000	99.255	1	0.744	100.000	99.269
			99.255				99.269

Australian Bond			
PV+	99.255	Full price	99.262
PV-	99.269	Par	30,000,000
PVBP	0.007	Market value	29,778,597
AnnModDur	0.737	MoneyDur	73.163

MoneyDur	73.163	ChgYield	0.01
		Exp. Loss	(219,490)

Romanian Bond: Full price, assuming 1 bp YTM increase		Romanian Bond: Full price, assuming 1 bp YTM decrease	
YTM (periodic)	0.0426	**YTM (periodic)**	0.0424

Period	Time to Receipt	Cash Flow	PV+	Period	Time to Receipt	Cash Flow	PV-
1	0.744	4.625	4.484	1	0.744	4.625	4.484
2	1.744	4.625	4.300	2	1.744	4.625	4.302
3	2.744	4.625	4.125	3	2.744	4.625	4.127
4	3.744	4.625	3.956	4	3.744	4.625	3.959
5	4.744	4.625	3.794	5	4.744	4.625	3.798
…	…	…	…	…	…	…	…
29	28.744	4.625	1.394	29	28.744	4.625	1.402
30	29.744	104.625	30.251	30	29.744	104.625	30.424
			107.254				107.604

Romanian Bond			
PV+	107.254	Full price	107.429
PV-	107.604	Par	30,000,000
PVBP	0.175	MV	32,228,627
AnnModDur	16.248	MoneyDur	1745.518
Full price	107.429	MV	32,228,627
MoneyDur	1745.518	ChgYield	0.01
		Exp. Loss	(5,236,553)

2. If the client believes that interest rates will rise in the future, with all else held constant, which bond investment would *most likely* perform best?

Hide Solution

Solution:

The Australian bond would perform best. Since it has the lowest duration and PVBP, the expected loss on the investment from an increase in its yield is lower than the BRWA and Romanian bonds. The expected loss from a 100 bp increase in yield on the BRWA bond, for example, is almost six times that for the Australian bond (USD219,490 versus USD1.3 million).

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Discussion

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MO Regarding practice questions which follow. So, C, technically is correct, but it is incorrect because people may get confused by terminology? I hope they don't put two correct answers on the actual exam and then say that the only reason the other correct answer is incorrect is because people may get confused....

Which of these definitions of duration is most relevant to a bond investor? A bond's duration is its:

A. half-life.

B. price sensitivity to yield changes.

C. first derivative of value with respect to its yield.

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to do with maturity (e.g., an interest only strip security.)

B. Correct. Bond investors are concerned about interest rate risk, and duration is a good measure of interest rate risk.

C. Incorrect because while duration was first measured using the calculus, referring to it in this way is confusing for investors who are unfamiliar with calculus or might confuse it with a derivative security.

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MO

Simple way of calculating PVBP, instead of calculating PV+ and PV-...

BRWA:

PVBP ≈ Price * ModDur * 0.0001

PVBP ≈ 100.815 * 4.335 * 0.0001 = 0.0437033

Australian:

PVBP ≈ Price * ModDur * 0.0001

PVBP ≈ 99.262 * 0.737 * 0.0001 = 0.00731561

Romanian:

PVBP ≈ Price * ModDur * 0.0001

PVBP ≈ 107.429 * 16.248 * 0.0001 = 0.17455064

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AS

if money duration is unitless because it is the rate of change of the position with respect to the change in yield per period then why modified duration is being mentioned in years? Is there any difference in in industry scenario?

Created 5 months ago by Ajay Saxena 1 reply | Last Activity: 5 months ago

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